

RCV Models with Demographic Controls (Imputed Data)

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```
library(mice)
library(dplyr)
library(lme4)
library(broom)
library(broom.mixed)
library(knitr)
library(kableExtra)
library(ggplot2)
library(scales)

load("../data/model_results.RData")
```

Overview

All models use a **quasibinomial GLM** with logit link on `cbind(rcv_yes, rcv_no)` — each ballot is treated as a Bernoulli trial, with quasibinomial correction for overdispersion. Results are pooled across 5 multiply-imputed datasets using Rubin's rules. Imputation excluded all RCV outcome variables.

Models:

- **M1:** Quasibinomial baseline — `dem_share * recent_lpw` (with interaction)
- **M2:** Quasibinomial + demographic controls (with interaction)
- **M3:** GLMM with random intercept by locale + controls (with interaction)

Demographic controls (PL94-171 + ACS 5-year): `pct_white_vap`, `pct_black_vap`, `pct_hispanic_vap`, `median_income`, `pct_bach_plus`, `pct_renter`

Data summary

```
diag_data %>%
  group_by(locale, rcv_jurisdiction, recent_lpw) %>%
  summarise(
    n = n(),
    mean_yes = mean(yes_share),
    mean_dem = mean(dem_share),
    mean_white = mean(pct_white_vap),
    med_income = median(median_income),
    mean_bach = mean(pct_bach_plus),
    .groups = "drop"
  ) %>%
  kable(digits = 3, caption = "Analysis sample by locale")
```

Table 1: Analysis sample by locale

locale	rcv_jurisdiction	recent_lpw	n	mean_yes	mean_dem	mean_white	mean_income	mean_bach
Alaska	statewide	1	578	0.486	0.403	0.534	5.952	0.197
Albany	local	0	3	0.733	0.892	0.489	10.313	0.728
Bloomington	local	0	32	0.512	0.638	0.754	7.822	0.428
Boulder	local	0	88	0.782	0.864	0.830	7.228	0.767
Eureka	local	0	8	0.603	0.687	0.721	4.320	0.290
Maine	statewide	1	527	0.456	0.376	0.968	5.343	0.262
Massachusetts	statewide	0	2173	0.470	0.673	0.715	8.046	0.442
Minnetonka	local	0	23	0.547	0.662	0.867	10.036	0.613

Model 1: Quasibinomial baseline

$$\text{logit}(p) = \beta_0 + \beta_1 \cdot \text{dem_share} + \beta_2 \cdot \text{recent_lpw} + \beta_3 \cdot \text{dem_share} \times \text{recent_lpw}$$

```
summary(pool_m1) %>%
  kable(digits = 4, caption = "M1: Quasibinomial baseline (pooled)")
```

Table 2: M1: Quasibinomial baseline (pooled)

term	estimate	std.error	statistic	df	p.value
(Intercept)	-2.7659	0.0239	-115.8576	3426.002	0
dem_share	3.9461	0.0355	111.1400	3426.002	0
recent_lpw	1.6124	0.0350	46.1332	3426.002	0
dem_share:recent_lpw	-1.2473	0.0647	-19.2909	3426.002	0

Model 2: Quasibinomial with demographic controls

$$\text{logit}(p) = \beta_0 + \beta_1 \cdot \text{dem_share} + \beta_2 \cdot \text{recent_lpw} + \beta_3 \cdot \text{dem_share} \times \text{recent_lpw} + \gamma \cdot \mathbf{X}_{\text{demo}}$$

```
summary(pool_m2) %>%
  kable(digits = 4, caption = "M2: Quasibinomial + controls (pooled)")
```

Table 3: M2: Quasibinomial + controls (pooled)

term	estimate	std.error	statistic	df	p.value
(Intercept)	-2.4423	0.0554	-44.0526	2771.6807	0e+00
dem_share	3.6693	0.0575	63.7827	917.5876	0e+00
recent_lpw	1.5855	0.0346	45.7759	2548.4716	0e+00
pct_white_vap	-0.3047	0.0387	-7.8750	2437.2629	0e+00
pct_black_vap	-1.3530	0.0637	-21.2434	1415.0516	0e+00
pct_hispanic_vap	-0.2053	0.0512	-4.0065	2769.7023	1e-04
median_income	-0.0096	0.0023	-4.1305	3054.2573	0e+00
pct_bach_plus	0.3454	0.0481	7.1783	1505.1735	0e+00
pct_renter	0.3226	0.0336	9.5979	2342.2407	0e+00
dem_share:recent_lpw	-1.2878	0.0631	-20.4161	2038.1661	0e+00

Adjusted pseudo-R² comparison

Deviance-based: $1 - \frac{\text{deviance}/\text{df}_{\text{resid}}}{\text{null deviance}/\text{df}_{\text{null}}}$. Penalizes for number of parameters.

```
tibble(
  Model = c("M1: Baseline", "M2: + Demographics"),
  `Pseudo-R2` = round(c(pr2_m1$pseudo_r2, pr2_m2$pseudo_r2), 4),
  `Adj. Pseudo-R2` = round(c(pr2_m1$adj_pseudo_r2, pr2_m2$adj_pseudo_r2), 4)
) %>%
  kable(caption = "Deviance-based pseudo-R2 (averaged across 5 imputations)")
```

Table 4: Deviance-based pseudo-R² (averaged across 5 imputations)

Model	Pseudo-R ²	Adj. Pseudo-R ²
M1: Baseline	0.8290	0.8289
M2: + Demographics	0.8603	0.8599

F-test: are demographic controls jointly significant?

```
f_tests <- lapply(1:length(fit_m1$analyses), function(i) {
  anova(fit_m1$analyses[[i]], fit_m2$analyses[[i]], test = "F")
})

tibble(
  Imputation = 1:length(f_tests),
  `F statistic` = sapply(f_tests, function(a) round(a$F[2], 1)),
  `p-value` = sapply(f_tests, function(a) format.pval(a$`Pr(>F)`[2], digits = 3))
) %>%
  kable(caption = "F-test: M1 vs M2 (demographic controls jointly)")
```

Table 5: F-test: M1 vs M2 (demographic controls jointly)

Imputation	F statistic	p-value
1	126.7	<2e-16
2	124.1	<2e-16
3	125.3	<2e-16
4	125.2	<2e-16
5	126.7	<2e-16

Overdispersion

```
tibble(
  Model = c("M1: Baseline", "M2: + Controls"),
  `Dispersion (dev/df)` = c(
    round(mean(sapply(fit_m1$analyses, function(m) deviance(m)/df.residual(m))), 1),
    round(mean(sapply(fit_m2$analyses, function(m) deviance(m)/df.residual(m))), 1)
  )
)
```

```
) %>%
  kable(caption = "Overdispersion (1.0 = none; quasibinomial corrects SEs)")
```

Table 6: Overdispersion (1.0 = none; quasibinomial corrects SEs)

Model	Dispersion (dev/df)
M1: Baseline	15.6
M2: + Controls	12.8

Interpretation (M2)

All coefficients are on the **log-odds scale**. The core finding holds: **dem_share** and **recent_lpw** remain strong positive predictors of RCV support. Among the demographic controls:

- **pct_bach_plus** (education) is the strongest new predictor — more educated precincts support RCV more, controlling for partisanship.
- **pct_renter** is positive — more urban/transient precincts favor RCV.
- **pct_black_vap** is strongly negative — precincts with more Black residents are less supportive, holding partisanship constant.
- **median_income** is negative — wealthier precincts slightly less supportive, holding education constant.

Model 3: GLMM with random intercept by locale

Accounts for unobserved locale-level variation (campaign spending, ballot language, media environment) via random intercepts.

```
summary(pool_m3) %>%
  kable(digits = 4, caption = "M3: GLMM random intercept + controls (pooled)")
```

Table 7: M3: GLMM random intercept + controls (pooled)

term	estimate	std.error	statistic	df	p.value
(Intercept)	-2.0887	0.0709	-29.4510	3395.2145	0
dem_share	3.6354	0.0203	179.2247	24.9781	0
recent_lpw	1.2379	0.1377	8.9914	3410.3807	0
pct_white_vap	-0.3976	0.0133	-29.7980	847.9345	0
pct_black_vap	-1.3768	0.0208	-66.2249	119.9238	0
pct_hispanic_vap	-0.2951	0.0161	-18.3356	207.3517	0
median_income	0.0060	0.0008	7.6213	63.9815	0
pct_bach_plus	0.0709	0.0148	4.8006	220.8839	0
pct_renter	0.3867	0.0124	31.0877	20.0537	0
dem_share:recent_lpw	-1.1887	0.0215	-55.3947	33.2586	0

Random effects (imputation 1)

```

ranef(fit_m3$analyses[[1]])$locale %>%
  tibble::rownames_to_column("locale") %>%
  rename(intercept_shift = `(Intercept)`) %>%
  arrange(desc(intercept_shift)) %>%
  kable(digits = 3,
        caption = "Locale random intercepts (log-odds, imputation 1)")

```

Table 8: Locale random intercepts (log-odds, imputation 1)

locale	intercept_shift
Boulder	0.276
Eureka	0.118
Minnetonka	0.056
Maine	0.046
Bloomington	0.033
Alaska	-0.046
Albany	-0.175
Massachusetts	-0.310

Model comparison

```

tibble(
  Model = c("M1: Quasibinom baseline",
            "M2: Quasibinom + demographics",
            "M3: GLMM + demographics"),
  `Adj. pseudo-R2` = c(
    round(pr2_m1$adj_pseudo_r2, 4),
    round(pr2_m2$adj_pseudo_r2, 4),
    "_"
  ),
  Notes = c(
    "",
    "Demographics jointly significant (F-test p < 2e-16)",
    "Random intercept absorbs locale-level heterogeneity"
  )
) %>%
  kable(caption = "Model overview")

```

Table 9: Model overview

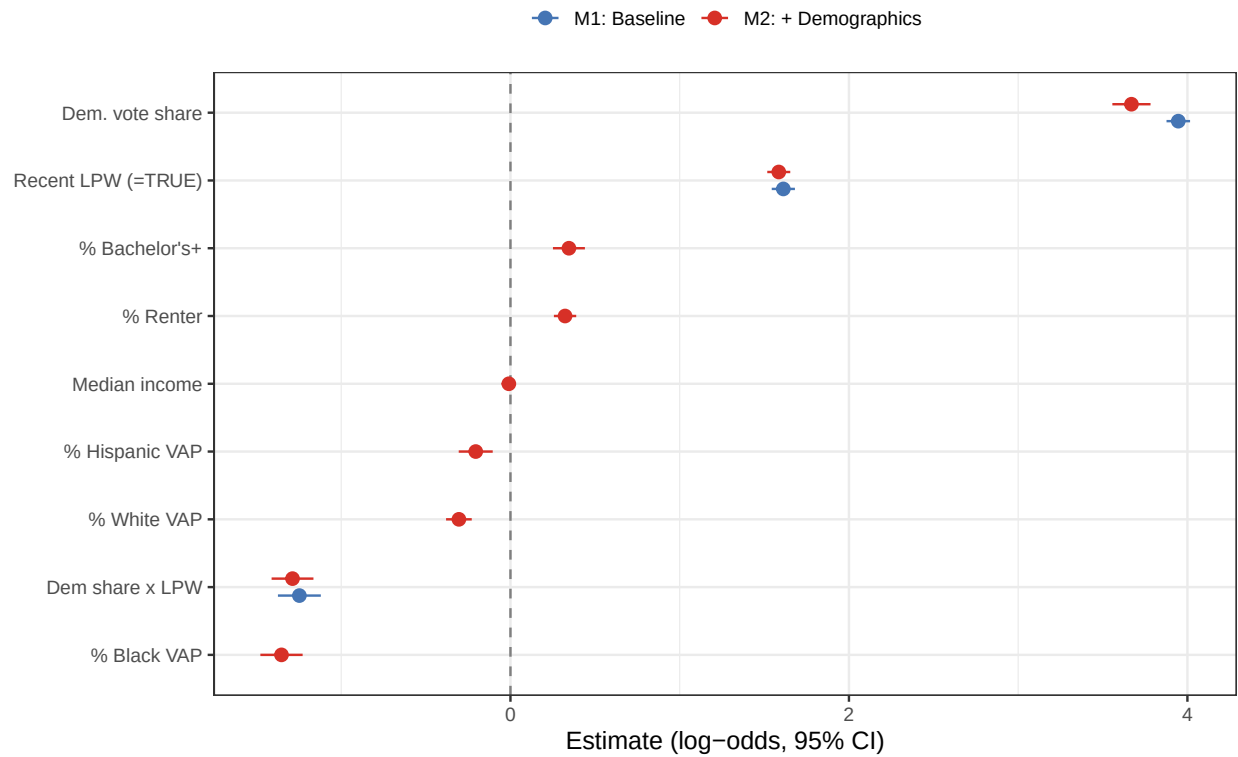
Model	Adj. pseudo-R ²	Notes
M1: Quasibinom baseline	0.8289	
M2: Quasibinom + demographics	0.8599	Demographics jointly significant (F-test p < 2e-16)
M3: GLMM + demographics	—	Random intercept absorbs locale-level heterogeneity

Coefficient comparison: M1 vs M2

```
coefs <- bind_rows(
  summary(pool_m1) %>% mutate(model = "M1: Baseline"),
  summary(pool_m2) %>% mutate(model = "M2: + Demographics")
) %>%
  filter(term != "(Intercept)") %>%
  mutate(
    lo = estimate - 1.96 * std.error,
    hi = estimate + 1.96 * std.error,
    term = case_match(term,
      "dem_share" ~ "Dem. vote share",
      "recent_lpw" ~ "Recent LPW (=TRUE)",
      "dem_share:recent_lpw" ~ "Dem share x LPW",
      "pct_white_vap" ~ "% White VAP",
      "pct_black_vap" ~ "% Black VAP",
      "pct_hispanic_vap" ~ "% Hispanic VAP",
      "median_income" ~ "Median income",
      "pct_bach_plus" ~ "% Bachelor's+",
      "pct_renter" ~ "% Renter",
      .default = term
    )
  )

ggplot(coefs, aes(x = estimate, y = reorder(term, estimate), color = model)) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "grey50") +
  geom_pointrange(aes(xmin = lo, xmax = hi),
    position = position_dodge(width = 0.5), size = 0.5) +
  scale_color_manual(values = c("M1: Baseline" = "#4575B4",
    "M2: + Demographics" = "#D73027")) +
  labs(x = "Estimate (log-odds, 95% CI)", y = NULL, color = NULL,
    title = "Quasibinomial coefficient comparison") +
  theme_bw(base_size = 13) +
  theme(legend.position = "top")
```

Quasibinomial coefficient comparison



Coefficient comparison: M2 vs M3

```
coefs_all <- bind_rows(
  summary(pool_m2) %>% mutate(model = "M2: Quasibinom"),
  summary(pool_m3) %>% mutate(model = "M3: GLMM")
) %>%
  filter(term != "(Intercept)") %>%
  mutate(
    lo = estimate - 1.96 * std.error,
    hi = estimate + 1.96 * std.error,
    term = case_match(term,
      "dem_share" ~ "Dem. vote share",
      "recent_lpw" ~ "Recent LPW (=TRUE)",
      "dem_share:recent_lpw" ~ "Dem share x LPW",
      "pct_white_vap" ~ "% White VAP",
      "pct_black_vap" ~ "% Black VAP",
      "pct_hispanic_vap" ~ "% Hispanic VAP",
      "median_income" ~ "Median income",
      "pct_bach_plus" ~ "% Bachelor's+",
      "pct_renter" ~ "% Renter",
      .default = term
    )
  )

ggplot(coefs_all, aes(x = estimate, y = reorder(term, estimate), color = model)) +
```

```
geom_vline(xintercept = 0, linetype = "dashed", color = "grey50") +
geom_pointrange(aes(xmin = lo, xmax = hi),
  position = position_dodge(width = 0.5), size = 0.5) +
scale_color_manual(values = c("M2: Quasibinom" = "#4575B4",
  "M3: GLMM" = "#1B7837")) +
labs(x = "Estimate (log-odds, 95% CI)", y = NULL, color = NULL,
  title = "Quasibinomial vs. GLMM coefficient comparison") +
theme_bw(base_size = 13) +
theme(legend.position = "top")
```

Quasibinomial vs. GLMM coefficient comparison

